Embeddings & pgvector: AI to Data Engineering

SkillForge · AI-11 handoff for DE-02 / DE-05 · 26 October 2026 · From: Ziad Abdullah (AI Lead)

At a glance

Item	Value
Embedding model	`BAAI/bge-small-en-v1.5` (ONNX, quantized, run locally)
Value stored in embedding_model	`local/BAAI/bge-small-en-v1.5`
Vector dimension	**384** (`vector(384)`)
Normalization	Unit length (L2-normalized) before writing
Distance metric	**Cosine**: HNSW with `vector_cosine_ops`
Search order / score	`ORDER BY embedding <=> query`; score = `1 - (embedding <=> query)`

1. The embedding_model value

Store this exact string, including the `local/` prefix, in the `embedding_model` column of every row:

```
local/BAAI/bge-small-en-v1.5
```

- If you check or filter on this column, match the whole string.
- The AI code refuses to mix vectors from different models or dimensions in one index, so any other value will be rejected.
- The environment variable `SKILLFORGE_MODEL_EMBEDDING` overrides the model. Please do not set it unless we agree to change models together.

2. Distance metric

Cosine. Vectors are normalized to unit length before they are written. Use an HNSW index with `vector_cosine_ops`. Search orders by `embedding <=> query`, and the similarity score is `1 - (embedding <=> query)`.

3. Table definition (DDL)

Generated from our code with `pg_schema("knowledge_chunks", 384)`:

```
CREATE EXTENSION IF NOT EXISTS vector;
CREATE TABLE IF NOT EXISTS knowledge_chunks (
    chunk_id        text PRIMARY KEY,
    source_url      text NOT NULL,
    title           text NOT NULL,
    retrieved       date NOT NULL,
    text            text NOT NULL,
    section         text[] NOT NULL DEFAULT '{}',
    concepts        text[] NOT NULL DEFAULT '{}',
    content_hash    char(64) NOT NULL,
    domain          text NOT NULL,
    embedding_model text NOT NULL,
    embedding       vector(384) NOT NULL
);
CREATE INDEX IF NOT EXISTS knowledge_chunks_embedding_idx
    ON knowledge_chunks USING hnsw (embedding vector_cosine_ops);
CREATE INDEX IF NOT EXISTS knowledge_chunks_source_idx
    ON knowledge_chunks (source_url);
CREATE INDEX IF NOT EXISTS knowledge_chunks_domain_idx
    ON knowledge_chunks (domain text_pattern_ops);
```

4. If you embed text yourselves

The easiest path is to call `ingest_pages()` from our code; it handles all of the rules below. If you embed in your own pipeline, you must match them exactly, or the vectors will not be comparable:

- **Passages / chunks:** embed the raw chunk text with **no prefix**.
- **Search queries only:** put this instruction before the query (note the trailing space):

```
Represent this sentence for searching relevant passages:
```

- **Same pinned model:** `Qdrant/bge-small-en-v1.5-onnx-Q` at revision `aa8f8b06`… (full pin in `ai/core/llm/models.yaml`). The model file is checked against its sha256 before every load.
- **Normalize** every vector to unit length before writing.

> **Why this matters:** a vector made with a different model, a different model version, or the wrong prefix lives in a different "space". Search results would be silently wrong, so our code checks `embedding_model` and the dimension and refuses to mix them.

Related docs (in the AI folder)

- `ai/docs/ai/retrieval.md`: retrieval, ingest, store interface, pgvector notes
- `ai/docs/ai/extraction.md`: how pages are cleaned and chunked (AI-10, used by DE-05)

Questions or changes to any of the above: contact Ziad. Changing the model or dimension is a contract change that both AI and DE must approve, because it requires rebuilding the index.